

Nov 06, 2019

Version 3

# Transient transformation of *Ostreococcus* species (OTTH595, RCC809 and RCC802) and *Bathycoccus* V.3

 [Nature Methods](#)

DOI

<https://dx.doi.org/10.17504/protocols.io.83uhynw>

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Protist Research to Opti...



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**DOI:** <https://dx.doi.org/10.17504/protocols.io.83uhynw>

**Protocol Citation:** francois-yves bouget 2019. Transient transformation of *Ostreococcus* species (OTTH595, RCC809 and RCC802) and *Bathycoccus*. [protocols.io https://dx.doi.org/10.17504/protocols.io.83uhynw](https://dx.doi.org/10.17504/protocols.io.83uhynw)

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**Protocol status:** Working

**We use this protocol and it's working**

**Created:** November 06, 2019

**Last Modified:** May 18, 2026

**Protocol Integer ID:** 29524

**Keywords:** *ostreococcus* species, *bathycoccus* this protocol, *bathycoccus*, transient gene expression, cells by electroporation, introduction of dna, measure of transient gene expression, rcc809, gene expression, preparation of cell, transient transformation, dna, electroporation, cell, transformation, selection of stable transformant



## Abstract

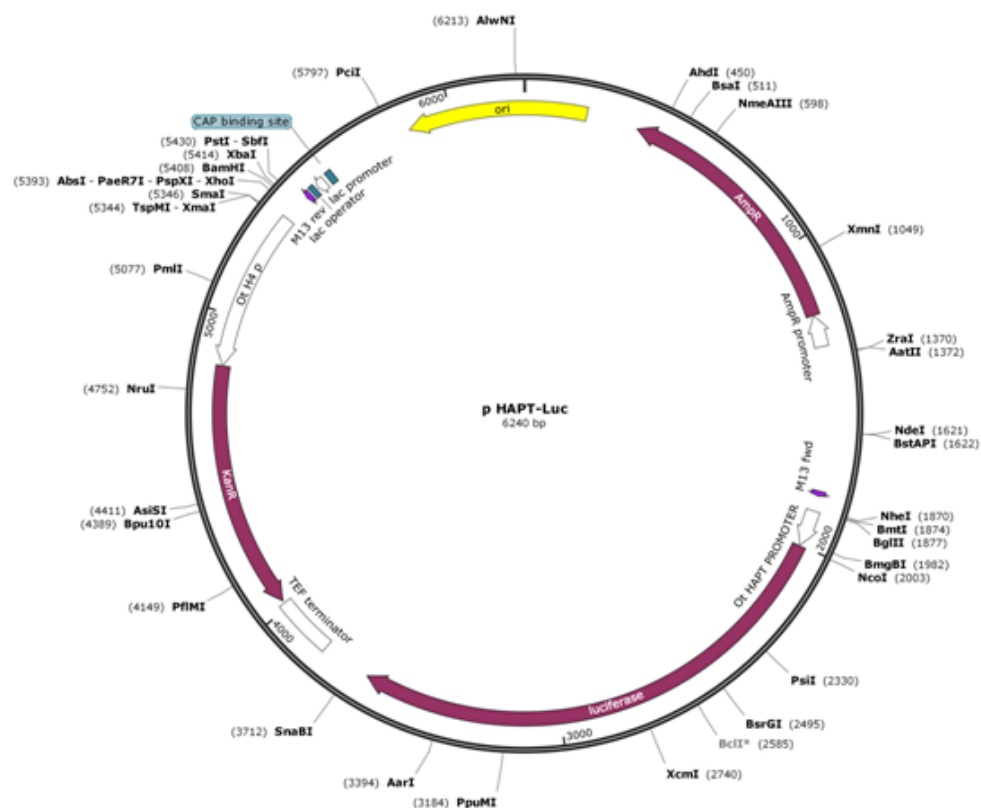
This protocol describes the preparation of cells and introduction of DNA into the cells by electroporation. For selection of stable transformants or measure of transient gene expression see related protocols.

## Materials

### *Ostreococcus lucimarinus* (RCC802)

pHAPT:Luc vector (Djouani Tahri et al., PLOS ONE 2011 <https://doi.org/10.1371/journal.pone.0028471>)

### pHAPT:Luc map



### PHAPT:Luc sequence

O. tauri pHAPT:Luc6240 bp ds-DNAcircular

DEFINITIONsynthetic circular DNA

FEATURESLocation/Qualifiers



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/mol_type="other DNA"
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```

Validation of *O. lucimarinus* transgenic lines by PCR (Figure 4A)

Forward primer: 1GAGCGCAACGGTACCCGGGCGGTACCTGTGCG

Reverse primer: CTGGCGACGCTGACGGCGTTACTTCACGT

### **Bathycoccus prasinus (RCC4222)**

**pH4:KanMx pHAPT:Luc** transgene (PCR product)

Construction of the transgene by PCR :

The transgene was generated by fusion PCR as described in Shevchuk et al. (2004) using DNA templates described below and oligonucleotides mapped on the sequence below.

**Promoters of Histone H4 and high affinity phosphate (HAPT) transporters** were amplified using (Fas1/Ras1) and (Fas4/Ras4) oligonucleotides respectively. KanMx and Luciferase sequences were amplified from pHAPT:luc plasmid template using oligonucleotides (Fas2/Ras2) and (Fas3/Ras3). Final PCR was done on amplified fragments using (Fas5/Ras5) oligonucleotides.

### **Oligonucleotide sequences:**

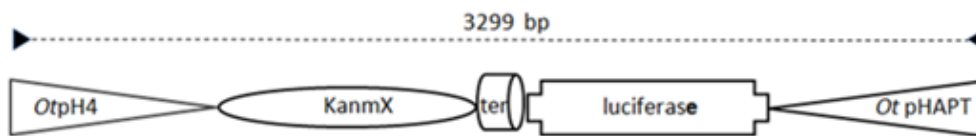
Fas1:CCGGCTTCGTGATGCCTTGGATGTTGTCTC



Ras1: TCGAAACGTGAGTCTTTTCCTTACCcatTGTGTTTGATTTATAATGAGGTTTTCTT  
 Fas2: AAGAAAACCTCATTATAAATCAAACACAatgGGTAAGGAAAAGACTCACGTTTCGA  
 Ras2: CCAAGAAGGGCGGAAAGATCGCCGTGTAAGAAATACCGTCTATCATCGATGAATTCGA  
 Fas3: CGAATTCATCGATGATAGACGGTATTTCTTACACGGCGATCTTTCCGCCCTTCTTGG  
 Ras3: AAGCATATATACATACACACACATACAAAATGGAAGACGCCAAAAACATAAAGAAAGG  
 Fas4: CCTTTCTTTATGTTTTTGGCGTCTTCCATTTTGTATGTGTGTGTATGTATATATGCTT  
 Ras4: AAGTGACACGACCAAACGCGCGTCGAC  
 Fas5: CCACCTTTACCTCTGCCGGACATTGTGA  
 Ras5: GCGCGGTAATATCTACGAGGTAGCACGAG

Note that transformants were checked by PCR using ( to amplify Kanx

#### Map of the transgene:



#### Sequence of the transgene:

4020 bp ds-DNAlinear

REFERENCE1(bases 1 to 4020)

AUTHORSFY Bouget and JC Lozano

FEATURESLocation/Qualifiers

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ORIGIN

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1141 tgattcaggt gaaaatattg ttgatgcgct ggcagtgttc ctgcgccggt tgcattcgat
1201 tctgtttgt aattgtcctt ttaacagcga tcgctgattt cgtctcgctc aggcgcaatc
1261 acgaatgaat aacggtttgg ttgatgcgag tgattttgat gacgagcgta atggctggcc
1321 tgttgaacaa gtctggaaag aatgcataa gctttgcca ttctaccggg attcagtcgt
1381 cactcatggt gatttctcac ttgataacct tattttgac gaggggaaat taatagggtg
1441 tattgatgtt ggacgagtcg gaatcgcaga ccgataccag gatcttgcca tcctatggaa
1501 ctgcctcggg gagttttctc cttcattaca gaaacggctt ttcaaaaat atggtattga
1561 taatcctgat atgaataaat tgcagtttca tttgatgctc gatgagtttt tctaataatc
1621 agtactgaca ataaaaagat tctgttttc aagaactgt catttgata gttttttat
1681 attgtagttg ttctatttta atcaaattgt agcgtgattt atatttttt tcgcctcgac
1741 atcatctgcc cagatgcgaa gttaagtgcg cagaaagtaa tatcatgcgt caatcgtatg
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1861 acgagctcga attcatcgat gatagacggt atttctaca cggcgatctt tccgcccttc
1921 ttggccttta tgaggatctc tctgattttt cttgcgtcga gttttccggt aagaccttc
1981 ggtacttcgt ccacaaacac aactcctcgc cgcaactttt tcgcggttgt tacttgactg
2041 gcgacgtaat ccacgatctc ttttccgctc atcgtctttc cgtgctccaa aacaacaacg
2101 gcggcgggaa gttcaccggc gtcacgtcgc ggaagacctg cgacacctgc gtcgaagatg
2161 ttggggtgtt ggagcaagat ggattccaat tcagcgggag ccacctgata gcctttgtac
2221 ttaatcagag acttcaggcg gtcaacgatg aagaagtgtt cgtcttcgct ccagtaagct
2281 atgtctccag aatgtagcca tccatccttg tcaatcaagg cgttggtcgc ttccggattg
2341 ttacataac cggacataat cataggacct ctcacacaca gttcgctctt ttgattaacg
```

```
2401 cccagcggtt tcccgggtatc cagatccaca accttcgctt caaaaaatgg aacaacttta
2461 ccgaccgcgc ccggtttatc atccccctcg ggtgtaatca gaatagctga tgtagtctca
2521 gtgagcccat atccttgctt gatacctggc agatggaacc tcttggaac cgcttccccg
2581 acttccttag agagggggagc gccaccagaa gcaatttcgt gtaaattaga taaatcgat
2641 ttgtcaatca gagtgccttt ggcaagaag gagaataggg ttggcaccag cagcgcactt
2701 tgaatcttgt aatcctgaag gctcctcaga aacagctctt ctcaaactt atacattaag
2761 acgactcgaa atccacatat caaatatccg agtgtagtaa acattccaaa accgtgatgg
2821 aatggaacaa cacttaaaat cgcagtatcc ggaatgattt gattgcaaaa aataggatct
2881 ctggcatgcg agaatctcac gcaggcagtt ctatgaggca gagcgacacc ttaggcaga
2941 ccagtagatc cagaggagtt catgatcagt gcaattgtct tgcctctatc gaaggactct
3001 ggcacaaaaat cgtattcatt aaaaccggga ggtagatgag atgtgacgaa cgtgtacatc
3061 gactgaaatc cctggtaatc cgttttagaa tccatgataa taatttttg gatgattggg
3121 agctttttt gcacgttcaa aatttttgc aaccctttt tggaacgaa caccacggta
3181 ggctgcgaaa tgccatact gttgagcaat tcacgttcat tataaatgtc gttcgcgggc
3241 gcaactgcaa ctccgataaa taacgcgccc aacaccggca taaagaattg aagagagttt
3301 tcaactgata cgacgattct gtgatttga ttcagcccat atcgtttcat agcttctgcc
3361 aaccgaacgg acatttcgaa gtactcagcg taagtgtatg ccacctcgat atgtgcatct
3421 gtaaaagcaa ttgtccagg aaccaggcg tctcttca tagccttatg cagttgctct
3481 ccagcgggtc catcttcag cggatagaat ggcgccgggc cttctttat gttttggcg
3541 tcttcattt tgtatgttg tgtatgtata tatgcttgg gaatatatgt tcacagaatg
3601 acgacttga aagcgcggtt gaattttta acgaaaatct ccgtgtggct gatattttt
3661 gcttttgct tttttcaac caccggatt ttgctttt tcaaaacaa cccaccgacc
3721 gtaaatgttg tgtgtcttg ttctgttg ggctgcttc ttttagagga gggaggatgc
3781 attcagagtt aatatattat atgtgctccg agatctgttg tatacgagga gttggtgtg
3841 gcttttaat acacaaaata cgcctaaacg cgaggaggcg tcgttgaaac gtaaaggta
3901 tattactcgt gctacctgt agatattacc gcgcgtaatt agaagtcgtg ggagttgtg
3961 tcgtgtcgt tgtatcgtc gacgcgcgtt tggctgtgta cacttactc gcgccgcgac
//
```

Validation of *B. prasinos* transgenic lines by PCR (Figure 4b)

## Oligonucleotide sequences

1(Fas5) CCACCTTTACCTCTGCCGGACATTGTGA

2TCGAAACGTGAGTCTTTTCCTTACC

3 TCGCCTCGACATCATCTGCCAGATGC



4 (Ras5)GCGCGGTAATATCTACGAGGTAGCACGAG

Reference:

Nikolai A. Shevchuk, Anton V. Bryksin, Yevgeniya A. Nusinovich, Felipe C. Cabello, Margaret Sutherland, Stephan Ladisch /Nucleic Acids Research/, Volume 32, Issue 2, 16 January 2004, Page e19,  
<https://doi.org/10.1093/nar/gnh014>

## Cell preparation

- 1
  - 1) Starting from a culture of *Ostreococcus tauri*, RCC809 or *Bathycoccus* in stationary phase, inoculate cultures at 1 million cells/ml as determined by flow cytometry (Accuri C6 BD) in 200 ml plastic flasks in Artificial Seawater supplemented with Keller medium supplement (trace metals, vitamins, nitrate and Phosphate as described in Djouani Tahri et al., PloS ONE 2011). For each transformation (including control), you should plan on using 50 ml de culture in exponential phase.
  - 2) Grow cells for 4 to 5 days depending on the light conditions, until they reach densities of  $30$  to  $40 \cdot 10^6$  cells/ml.
  - 3) Count cells by flow cytometry. Check by SyBR Green II staining that bacterial contamination is below 2%.
  - 4) Transfer cultures to 50 ml Falcon tubes.
  - 5) Centrifuge at 8000g for 10 min at 4°C.
  - 6) Remove the supernatant, resuspend the cell pellet in 1 ml de sorbitol 1M (pH 7.5) in H<sub>2</sub>O MQ, at 4°C.
  - 7) Transférer the cell suspension to 1.5 ml eppendorf.
  - 8) Centrifuge at 8000g for 10 min at 4°C.
  - 9) Remove 900 µl of supernatant
  - 10) **Resuspend cells by gently pipeting.**

## Electroporation of the transgene

- 2
  - 1) Add 5µg of transgene DNA to cell suspension. Keep on ice for 5 minutes. The transgene consist of the high affinity phosphate promoter fused to the firefly luciferase (see Djouani Tahri et al., PloS one 2011).
  - 2) Transfer cells to a 2 mm electroporation cuvette (Biorad).
  - 3) Apply an electric field  
For *Ostreococcus tauri* (OTTH595) : **capacitance:** 25µF, **resistance** 600 Ω, **voltage** 1.35KV.  
For *Ostreococcus* sp RCC809 : **capacitance:** 25µF, **resistance** 600 Ω, **voltage** 1.4KV.  
For *Bathycoccus* (RC4222) : **capacitance:** 25µF, **resistance** 600 Ω, **voltage** 1.5KV.  
For *Ostreococcus lucimarinus* RCC802 : **capacitance:** 25µF, **resistance** 600 Ω, **voltage** 1.2KV.
  - 4) Add 1ml of fresh culture Medium to resuspend the cells.
  - 5) Add 40 ml of culture medium and transfer to a culture flask.
  - 6) Incubate at 20°C overnight in a light incubator.

At this stage, transient transgene expression can be measured or stable transformants can be selected (see relevant protocols).

